

ENTERPRISE LAN

Designing a Secure and Fault Tolerant Enterprise Network

Final Technical Report

Preston Elia

Principal Investigator

May 2026

1. Executive Summary

This capstone designs and implements a secure, enterprise local area network (LAN) in a visualized environment. The network architecture uses PfSense firewall on a Proxmox Virtual Machine (VM) server which was used to create 4 main security zones which were User, Server, Management, and DMZ over VLAN tagged virtual links. Each zone has its own subnet, DHCP scope, and firewall policy, with inter zone traffic being controlled by explicit rules, otherwise dropped by the default deny implementation.

The implementation was tested through a 12 test matrix covering DHCP delivery, allow and block enforcement, inter vlan routing, default deny posture, and firewall logging. All twelve results were expected except one is labeled with NOTE rather than FAIL because it was an engineering finding that is integrated into PfSense so can admin not lose the ability to edit its firewall rules.

Three engineering findings emerged that are worth highlighting: (1) firewall ordering matters, when misordered ruleset was diagnosed and fixed while testing was being taken. (2) firewall as target as tests are not a reliable way to measure a cross zone enforcement. (3) PfSense default deny implements a zero trust posture without putting block rules. Documented in Section 7.

Two scope reductions were taken during the project and were Documented in Section 8. Production WAN connectivity (Air Gapped) and a high availability firewall.

2. Introduction

2.1 Background

Small to mid sized companies often deploy networks without standardized design principles which lead to bad security for example limited segmentation and brittle fault tolerance. A flat Layer 2 network is very easy to set up when you are trying to get a business up and running but it provides no internal trust boundaries. A compromise of host trivially exposes every other host on the same domain. Enterprise reference designs (Cisco's three tier model, NIST SP 800-41) consistently recommend segmenting networks into security zones with policy controlled boundaries between them.

This capstone applies those principles in a virtualized lab. The goal of this was not to reinvent the wheel but to show that mainstream open source tools (pfSense, Proxmox VE, Linux bridges with VLAN tagging) can be used together to make a defensible enterprise design that enforces segmentation.

2.2 Problem Statement

How can a small to mid sized enterprise build a network that (a) segments user, server, management, and public facing zones; (b) enforces inter zone access through firewall policy rather than network topology alone; and (c) provides per zone services like DHCP using only commodity open source software on a single hypervisor host?

2.3 Differentiation

This design emphasizes:

- Layered network architecture rather than a flat Layer 2 design.
- VLAN based segmentation enforced at the firewall, not just at the switch.
- Explicit firewall policy with most-specific-first rule ordering.
- Default-deny per interface, implementing zero trust between zones.
- Centralized DHCP per zone, with the firewall as the gateway and policy enforcement point.

3. Project Objectives and Scope

3.1 In Scope

- Design and implement a virtualized enterprise LAN topology with four security zones.
- Deploy pfSense as the firewall, router, and DHCP server.
- Implement firewall policy enforcing inter zone access controls.
- Validate the implementation through structured testing with documented results.
- Produce logical and physical topology diagrams suitable for use in operations documentation.
- Document the implementation, testing methodology, results, and findings.

3.2 Out of Scope

- Production WAN / internet connectivity. The lab is air gapped due to upstream gateway constraints. Outbound NAT is configured and structurally validated; live internet routing is not exercised.
- High availability firewall pair (CARP). Deferred because the lab uses a single Proxmox host. Service level redundancy is the in scope alternative.
- Centralized application services (DNS, web, monitoring) on Ubuntu Server. Originally planned. deferred to focus on core architectural deliverables.
- Hardware Layer 2/Layer 3 switching on a Cisco Catalyst 3750. The switch is on hand but not integrated. switching is simulated via a VLAN aware Proxmox bridge.

4. Network Architecture

4.1 Design Principles

The architecture follows three established design principles:

- **Segmentation.** Each VLAN is a separate broadcast domain and a separate trust zone. No host on one VLAN can talk to a host on another VLAN without traversing the firewall.
- **Defense in depth.** Multiple controls between an attacker and a target. Even if a host on the User LAN is compromised, it cannot reach the Management or DMZ zones without first defeating the firewall policy.
- **Least privilege.** Each zone gets only the access required for its role. The User LAN can reach the Server zone on web/SSH only, not on arbitrary ports. Management and DMZ are blocked entirely from the user zone.

4.2 Logical Topology

The logical topology is based around the pfSense firewall, with 4 security zones connected with VLAN tagged interfaces. Each zone has its own /24 subnet, DHCP scope, and firewall policy. The only way to get to a different zone is through the PfSense. There is no bypass like layer 2 path.

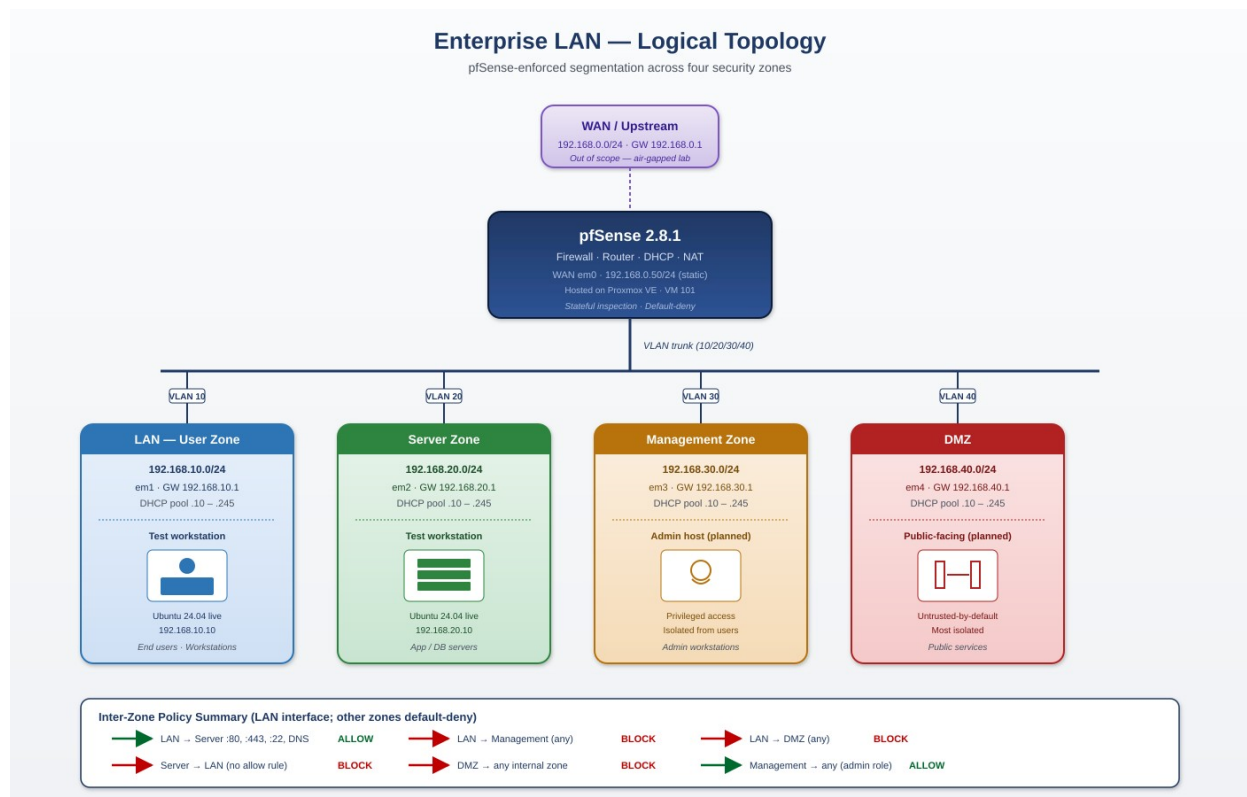


Figure 1: Logical topology — four security zones behind a single pfSense firewall.

4.3 Physical Topology

Physically, the entire network runs on a single Proxmox VE host. The pfSense firewall and all test workstations are virtual machines. The Proxmox VLAN aware Linux bridge (vmbro) functions as the enterprise managed switch, tagging traffic per VM and trunking VLAN IDs 10/20/30/40 to the pfSense VM's four LAN 2 side interfaces. Upstream connectivity is provided by a TP-Link TL-SG108E managed switch and a MoCA bridge to the lab's gateway.

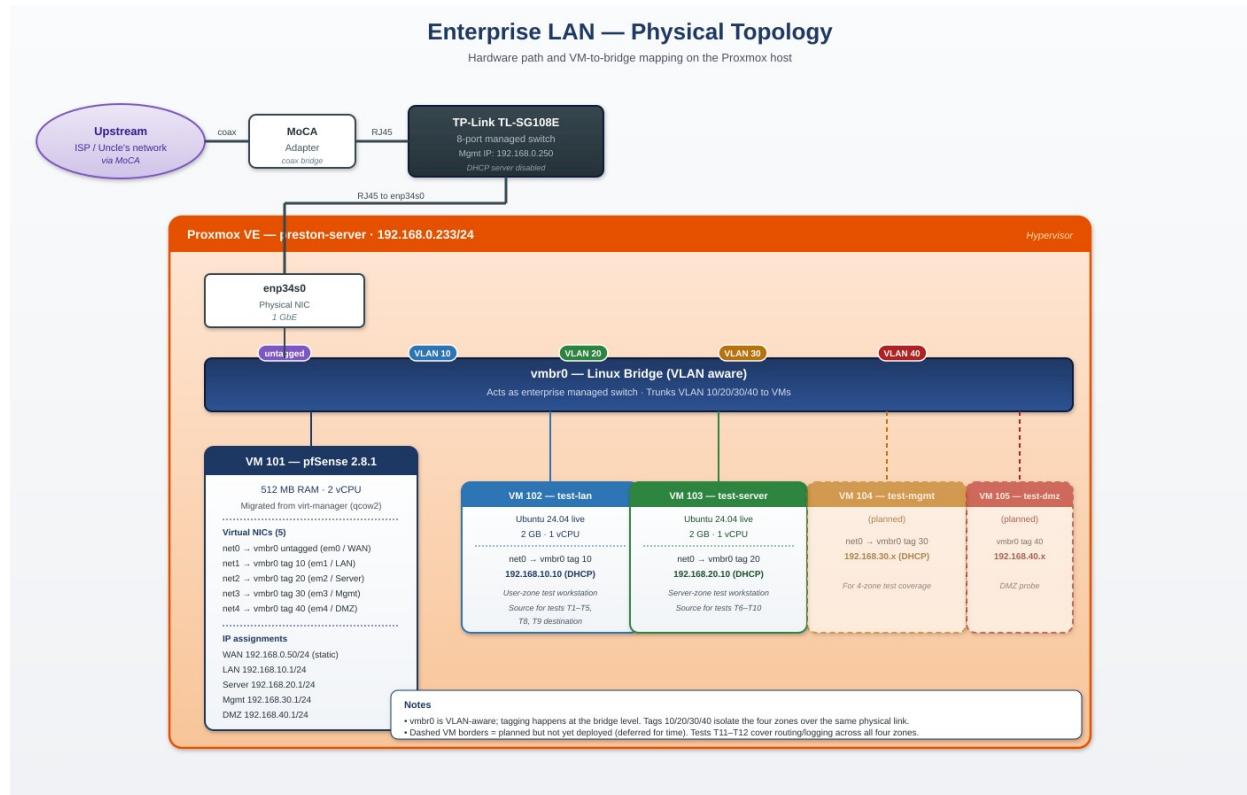


Figure 2: Physical topology — hardware path and VM to bridge mapping.

4.4 Zone Summary

Zone	Subnet	pfSense Interface / IP	DHCP Pool	Purpose
LAN (User)	192.168.10.0/24	em1 / 192.168.10.1	.10–.245	End user workstations
Server	192.168.20.0/24	em2 / 192.168.20.1	.10–.245	App / DB servers
Management	192.168.30.0/24	em3 / 192.168.30.1	.10–.245	Admin workstations
DMZ	192.168.40.0/24	em4 / 192.168.40.1	.10–.245	Public facing services
WAN (Lab)	192.168.0.0/24	em0 / 192.168.0.50 static	n/a	Upstream

5. Implementation

5.1 Hypervisor

The host is a dedicated Proxmox VE 8 server (192.168.0.233/24). All VMs and the virtual switch (vbr0) live on this host. Proxmox was chosen over the original libvirt/virt-manager setup after early implementation revealed limitations around VM management and network configuration; the migration is described in Section 5.5.

5.2 Firewall — pfSense

The firewall is pfSense 2.8.1 running as VM 101 with 512 MB RAM and 2 vCPUs. It has five virtual NICs:

- net0 → vbr0 untagged → em0 (WAN), 192.168.0.50/24 static.
- net1 → vbr0 tag 10 → em1 (LAN), 192.168.10.1/24.
- net2 → vbr0 tag 20 → em2 (Server), 192.168.20.1/24.
- net3 → vbr0 tag 30 → em3 (Management), 192.168.30.1/24.
- net4 → vbr0 tag 40 → em4 (DMZ), 192.168.40.1/24.

MAC addresses were preserved across the migration so that pfSense's interface assignments carried over without manual reassignment. WAN was set to a static IP after testing showed that the upstream lab gateway did not reliably issue DHCP leases for the locally administered MAC range used by QEMU/KVM virtual NICs.

5.3 DHCP

Since of the limitations of libvirt and having to scale down the project building a full Ubuntu DNS/DHCP server wasn't possible but PfSense has a build in DHCP called Kea. When Kea DHCP is enabled on each VLAN. Pools run from .10 to .245 in each /24 subnet so I had room to make some IPs static outside of the pool if needed. DNS registration is enabled so that DHCP leases are auto registered with the pfSense DNS Resolver, allowing test workstations to resolve each other by hostname.

5.4 Firewall Policy

The LAN interface ruleset is the most explicit in the design. Server, Management, and DMZ interfaces have no explicit allow rules they operate under the implicit default deny, which means hosts in those zones cannot initiate connections to other zones. This is the intended zero trust posture.

#	Action	Source	Destination	Port	Description
1	Pass	Anti Lockout (auto)	any → LAN address:443/80	*	Always allow LAN admin access
2	Pass	LAN subnets	192.168.20.0/24	TCP 80, 443	Web access to Server zone
3	Pass	LAN subnets	192.168.20.0/24	TCP 22	SSH to Server zone
4	Pass	LAN subnets	192.168.20.10	TCP/UDP 53	DNS to centralized resolver
5	Block	LAN subnets	192.168.30.0/24	*	Block users from Management
6	Block	LAN subnets	192.168.40.0/24	*	Block users from DMZ
7	Pass	LAN subnets	any	*	Catch all (future internet)
—	Block	any	any	*	Default deny (implicit)

Rules are evaluated top to bottom; the first matching rule wins. Rule ordering was an explicit finding during testing — see Section 7.1.

5.5 Migration from virt-manager to Proxmox

The pfSense VM was originally deployed under libvirt on my Linux mint laptop. During setup, limitation around VM management and VM connectivity I had to migrate what was already done to a self host Promox VM Server. The migration preserved the qcow2 disk image and all interface MAC addresses, allowing the pfSense interface to VLAN mapping to carry over without reconfiguration. The migration is documented as an implementation decision (engineering trade off) rather than a project pivot.

6. Testing and Validation

6.1 Test Methodology

Each test was executed from a live test workstation using built in tooling. ICMP reachability was tested with ping. TCP reachability was tested with bash's built in /dev/tcp network testing tools, which performs a TCP handshake without requiring extra utilities. DHCP delivery was verified with `ip a` after triggering Network Manager to reacquire a lease. All tests were captured as terminal screenshots and pfSense firewall log entries.

6.2 Test Environment

- Hypervisor: Proxmox VE on dedicated hardware.
- Firewall: pfSense 2.8.1 (VM 101).
- Test workstation 1: Ubuntu 24.04 live ISO on LAN VLAN, IP 192.168.10.10.
- Test workstation 2: Ubuntu 24.04 live ISO on Server VLAN, IP 192.168.20.10.
- Switching: Proxmox VLAN aware bridge (vmb0); VLAN tags 10/20/30/40.

6.3 Test Matrix

#	Source	Destination	Protocol	Expected	Actual	Result
T1	LAN VM (10.10)	Server gw (20.1)	TCP/443	Allow	OK	PASS
T2	LAN VM (10.10)	Server gw (20.1)	TCP/22	Allow*	Blocked	PASS
T3	LAN VM (10.10)	Mgmt gw (30.1)	TCP/443	Block	Blocked	PASS
T4	LAN VM (10.10)	DMZ gw (40.1)	TCP/443	Block	Blocked	PASS
T5	LAN VM (10.10)	DHCP (LAN)	DHCP	Lease	.10/24	PASS
T6	Server VM (20.10)	DHCP (Server)	DHCP	Lease	.10/24	PASS
T7	Server VM (20.10)	Server gw (20.1)	TCP/443	Allow	OK	PASS
T8	Server VM (20.10)	LAN host (10.10)	ICMP	Block	100% loss	PASS
T9	Server VM (20.10)	LAN host (10.10)	TCP/22	Block	Blocked	PASS
T10	Server VM (20.10)	LAN gw (10.1)	TCP/443	Block	OK	NOTE
T11	Inter VLAN routing	All gateways	L3	Reach	Reach	PASS

#	Source	Destination	Protocol	Expected	Actual	Result
T12	Firewall logging	System Logs	n/a	Captured	Captured	PASS

*Result legend: PASS = observed behavior matches the security policy. NOTE = observed behavior is not a failure but warrants explanation (see Section 7.2). * = T2 has an allow rule, but pfSense's SSH daemon is not enabled, so the connection is refused at the application layer; this is layered defense, not a policy failure.*

7. Engineering Findings

7.1 Rule order matters

When I was first setting up the firewall configuration on the LAN placed a 'allow internet access' rule above the block rules for the management and DMZ zones. Because Pfsense evaluates rules top down and stops at the first match, the allow rule stop the access to the rest of the block rules. I found this out because Test T3 and T4 returned OK instead of BLOCKED. The fix was just reorder the rules so the 'allow internet access' was on the bottom which with Pfsense web UI is just a drag and drop.

7.2 Default deny works as designed

The Server, Management, and DMZ interfaces have no explicit allow rules in their tabs, only the implicit default deny. Tests T8 and T9 confirmed that hosts in the Server zone cannot initiate connections to hosts in the LAN zone. Neither ICMP nor TCP. This is the desired zero trust posture. zones are silent until policy explicitly opens a path. It is also the simplest part of the design to maintain. as long as no allow rule is added, the zone remains isolated.

8. Limitations and Future Work

- WAN Connection in the lab was unable to work because of the configuration of the router given by the ISP which was unable to be manged for this lab. To work around this the lab was air gapped so there was no internet service but the computers on the network could communicate if on a VLAN that allowed them to.
- A true redundant high availability would have to have a second pfsense node but due to time constraints and migration problems the main focus was shifted to one pfsense node and completing SOW
- Testing was conducted from one workstation per zone (LAN and Server). Future work. expand to all four zones and longer duration testing under simulated load.

10. References

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